

Name Key

Physics 390 In-Class Activity 10

$$C = 1 - e^{-(\mu_o - \mu_B)x_o}$$

Here you will compare the contrast between soft tissue and a 1-mm-diameter artery that has been contrast medium-stained with an Omnipaque™ (iodine contrast) stain for photon energies slightly above, and slightly below, the iodine K absorption edge, which lies at 33 keV.

Use the following data:

$\mu = 0.6 \text{ cm}^{-1}$ for Omnipaque™ for energies slightly below the edge

$\mu = 2.0 \text{ cm}^{-1}$ for Omnipaque™ for energies slightly above the edge

$\mu = 0.4 \text{ cm}^{-1}$ for surrounding soft tissue at both energies

A) Compute the contrast for energies slightly below the edge.

$$C = 1 - e^{-(0.6 - 0.4) \times 0.1} = 0.02$$

B) Compute the contrast for energies slightly above the edge.

$$C = 1 - e^{-(2.0 - 0.4) \times 0.1} = 0.15$$

C) Do your results support the idea that contrast is significantly enhanced by using X-ray energies slightly above the edge energy?

Yes there is a 7.5-fold contrast increase above the edge energy